
CO5 ACTIVITY 1

COMPARATIVE ANALYSIS OF MACHINE LEARNING ALGORITHMS FOR LAPTOP BATTERY HEALTH PREDICTION USING ADVANCED MATPLOTLIB VISUALIZATIONS

NAME: JEYA HARSHINI R

REG NO: 192424051

1. Introduction

Battery health plays a significant role in determining the performance, reliability, and lifespan of modern laptops. Frequent charging cycles, overheating issues, excessive usage hours, and aging batteries can reduce battery efficiency over time. Predicting battery health helps users make informed decisions regarding battery maintenance and replacement.

Machine Learning techniques provide an effective solution for predicting battery health based on historical usage patterns and device characteristics. This project compares multiple machine learning algorithms to identify the most effective model for battery health prediction. Various visualization techniques using Matplotlib are employed to analyze data patterns and evaluate model performance.

```
(50, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 12 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   device_id                            50 non-null     object
 1   brand                                50 non-null     object
 2   model_year                           50 non-null     int64
 3   os                                    50 non-null     object
 4   usage_type                           50 non-null     object
 5   daily_usage_hours                    50 non-null     int64
 6   charging_cycles                      50 non-null     int64
 7   avg_charge_limit_percent             50 non-null     int64
 8   battery_health_percent               50 non-null     int64
 9   battery_age_months                  50 non-null     int64
10   overheating_issues                   50 non-null     object
11   performance_rating                   50 non-null     int64
dtypes: int64(7), object(5)
memory usage: 4.8+ KB
```

None	model_year	daily_usage_hours	charging_cycles	\
count	50.000000	50.000000	50.000000	
mean	2020.560000	6.600000	526.400000	
std	1.567007	1.498298	220.405563	
min	2018.000000	4.000000	180.000000	
25%	2019.250000	6.000000	342.500000	
50%	2021.000000	6.000000	495.000000	
75%	2022.000000	8.000000	695.000000	
max	2023.000000	10.000000	950.000000	
	avg_charge_limit_percent	battery_health_percent	battery_age_months	\
count	50.000000	50.000000	50.000000	
mean	92.100000	80.160000	42.240000	
std	6.857887	11.83278	18.118161	
min	80.000000	55.000000	12.000000	
25%	85.000000	72.500000	25.500000	
50%	90.000000	82.500000	38.000000	
75%	100.000000	90.000000	58.500000	
max	100.000000	98.000000	72.000000	
	performance_rating			
count	50.000000			
mean	3.580000			
std	1.144463			
min	1.000000			
25%	3.000000			
50%	4.000000			
75%	4.750000			
max	5.000000			

2. Objective

The main objectives of this project are:

- To analyze laptop battery health data.
- To preprocess and prepare the dataset for machine learning.
- To implement and compare multiple machine learning algorithms.
- To evaluate model performance using regression metrics.
- To visualize results using advanced Matplotlib functions.
- To identify the most accurate model for battery health prediction.

3. Dataset Description

The dataset contains information related to laptop battery usage and health.

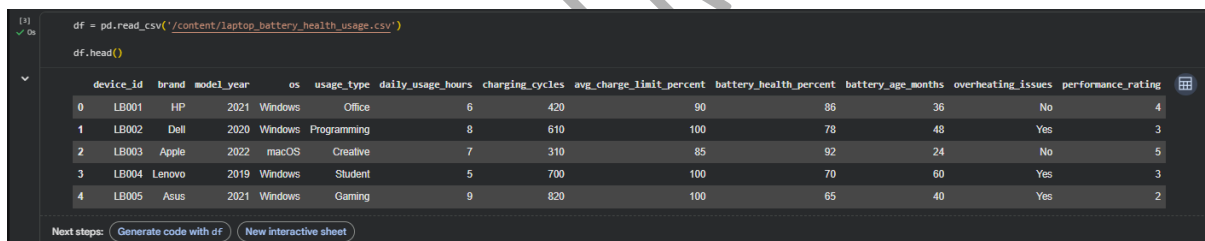
Dataset Attributes

Attribute	Description
Device ID	Unique identifier
Brand	Laptop manufacturer
Model Year	Manufacturing year

Attribute	Description
Operating System	OS used
Usage Type	Type of user activity
Daily Usage Hours	Average daily usage
Charging Cycles	Number of charge cycles
Battery Age Months	Age of battery
Average Charge Limit	Charging percentage limit
Overheating Issues	Presence of overheating
Performance Rating	Overall performance score
Battery Health Percent	Target variable

The target variable selected for prediction is:

Battery Health Percent



The screenshot shows a Jupyter Notebook interface. The code cell contains the following Python code:

```
df = pd.read_csv('/content/laptop_battery_health_usage.csv')
df.head()
```

The output displays the first five rows of the dataset as a table with 12 columns: device_id, brand, model_year, os, usage_type, daily_usage_hours, charging_cycles, avg_charge_limit_percent, battery_health_percent, battery_age_months, overheating_issues, and performance_rating.

	device_id	brand	model_year	os	usage_type	daily_usage_hours	charging_cycles	avg_charge_limit_percent	battery_health_percent	battery_age_months	overheating_issues	performance_rating
0	LB001	HP	2021	Windows	Office	6	420	90	86	36	No	4
1	LB002	Dell	2020	Windows	Programming	8	610	100	78	48	Yes	3
2	LB003	Apple	2022	macOS	Creative	7	310	85	92	24	No	5
3	LB004	Lenovo	2019	Windows	Student	5	700	100	70	60	Yes	3
4	LB005	Asus	2021	Windows	Gaming	9	820	100	65	40	Yes	2

At the bottom of the notebook, there are buttons for "Next steps: Generate code with df" and "New interactive sheet".

4. Data Preprocessing

The following preprocessing techniques were applied:

4.1 Handling Dataset

- Dataset loaded using Pandas.
- Data types verified.
- Statistical summary generated.

4.2 Encoding

Categorical features such as:

- Brand
- Operating System

- Usage Type
- Overheating Issues

were converted into numerical values using Label Encoding.

4.3 Feature Selection

Independent variables were separated from the target variable.

4.4 Data Splitting

The dataset was divided into:

- Training Data: 80%
 - Testing Data: 20%
-

5. Machine Learning Algorithms Used

5.1 Linear Regression

Linear Regression establishes a linear relationship between independent variables and battery health percentage.

5.2 Decision Tree Regressor

Decision Tree Regressor predicts battery health by recursively splitting data into smaller subsets.

5.3 Random Forest Regressor

Random Forest combines multiple decision trees and generates highly accurate predictions through ensemble learning.

5.4 Support Vector Regressor (SVR)

SVR uses support vectors to model complex relationships between features and target values.

5.5 K-Nearest Neighbors Regressor (KNN)

KNN predicts battery health based on the nearest neighboring observations.

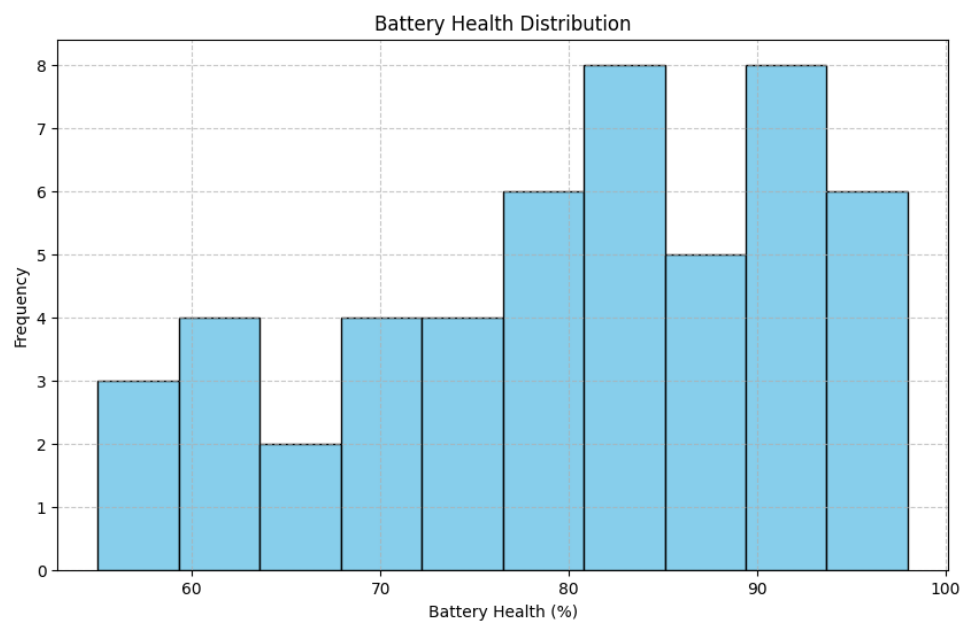
6. Data Visualization

Several Matplotlib visualizations were generated to analyze the dataset.

Visualization 1: Battery Health Distribution

Purpose:

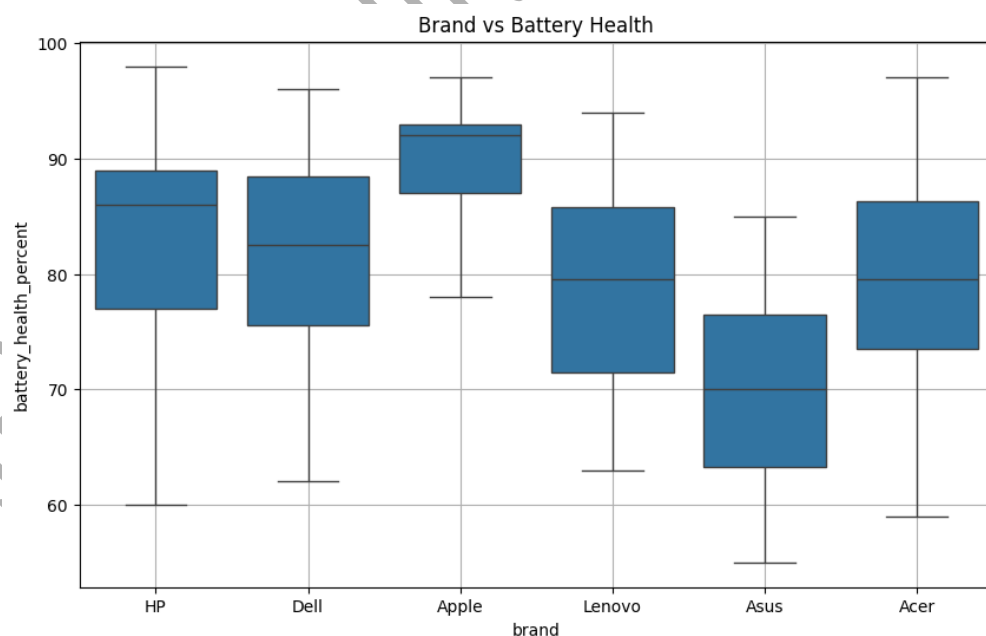
- Analyze battery health frequency distribution.



Visualization 2: Brand-wise Battery Health

Purpose:

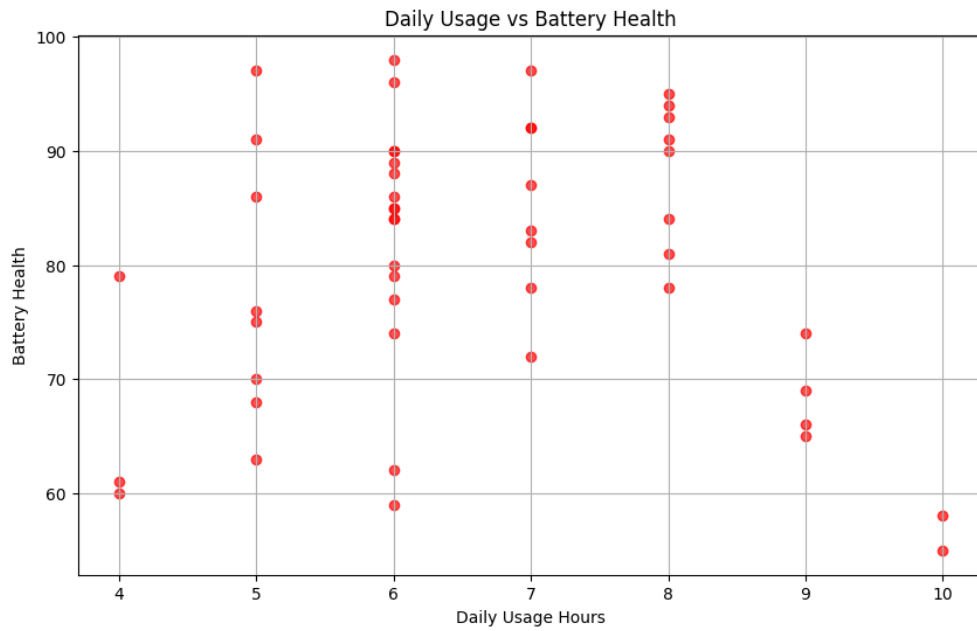
- Compare battery health among laptop brands.



Visualization 3: Daily Usage vs Battery Health

Purpose:

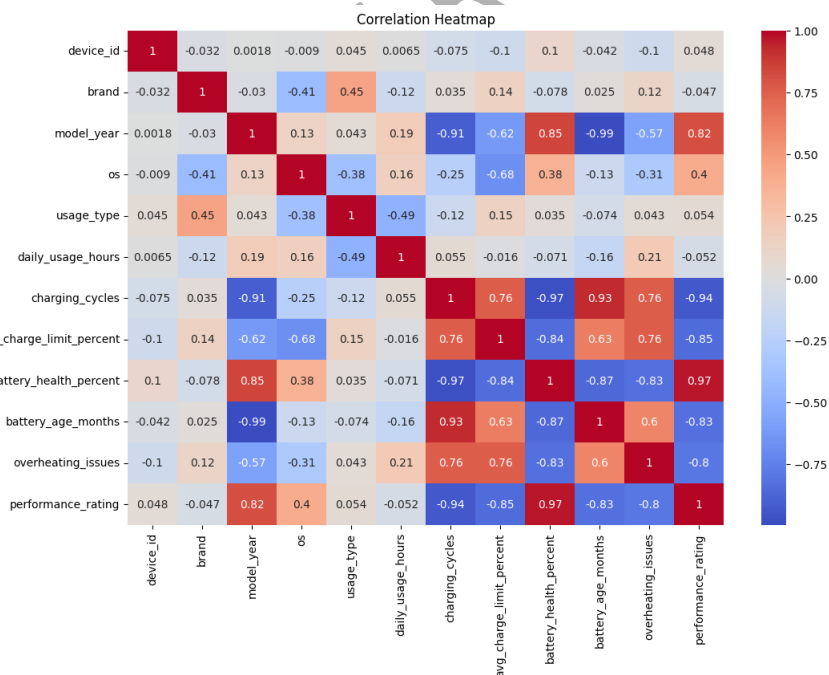
- Identify relationship between usage hours and battery condition.



Visualization 4: Correlation Heatmap

Purpose:

- Examine relationships among variables.



7. Model Evaluation Metrics

The following evaluation metrics were used:

Mean Absolute Error (MAE)

Measures average prediction error.

Root Mean Square Error (RMSE)

Measures standard deviation of prediction errors.

R-Squared Score (R^2)

Measures goodness of fit between predicted and actual values.

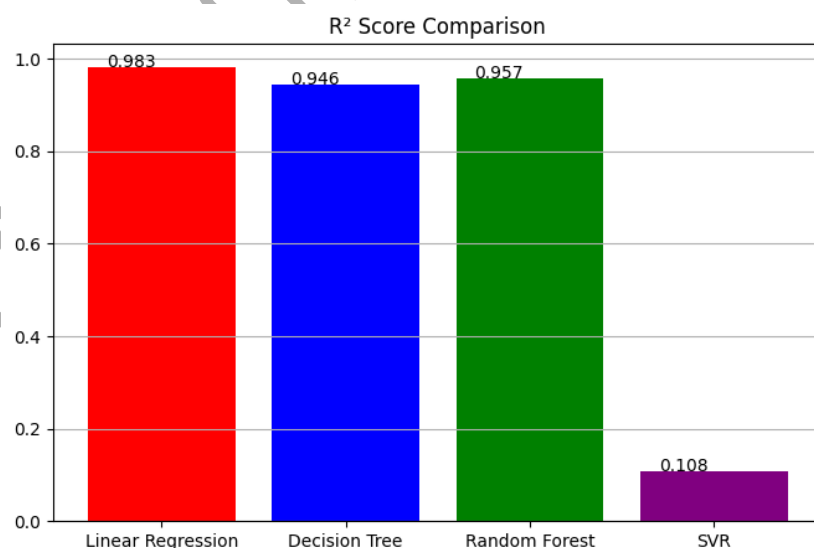
8. Performance Comparison

The machine learning models were evaluated using MAE, RMSE, and R^2 Score.

	Model	MAE	RMSE	R2
0	Linear Regression	1.237094	1.368078	0.983079
1	Decision Tree	2.200000	2.449490	0.945755
2	Random Forest	1.478000	2.170470	0.957409
3	SVR	8.928046	9.935733	0.107506

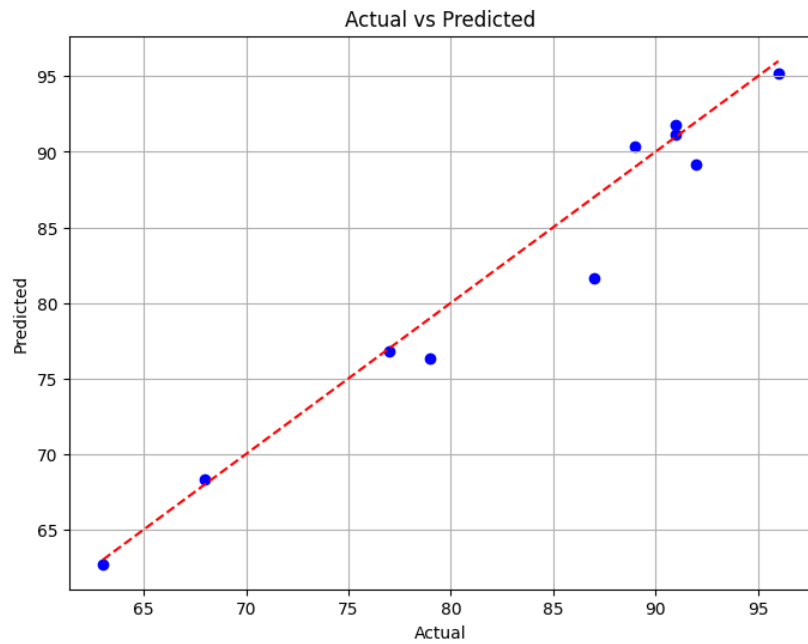
9. Accuracy Comparison Graph

The R^2 Score values of all models were compared visually.



10. Actual vs Predicted Analysis

Actual battery health values were compared with model predictions.

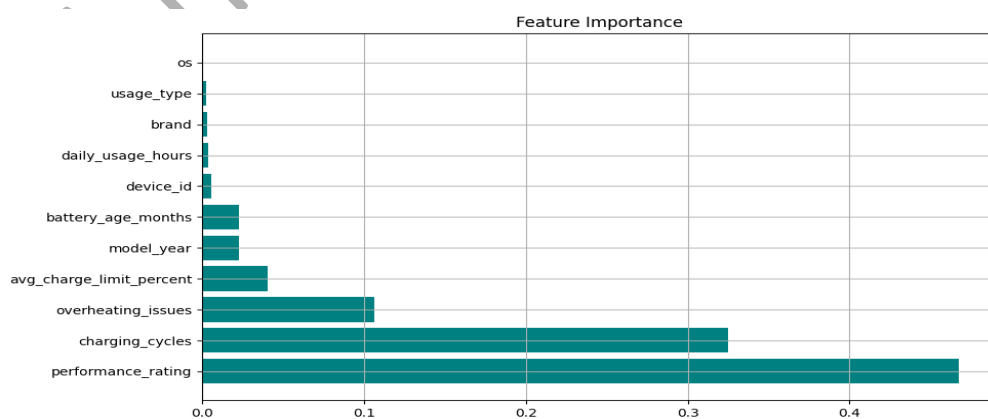


11. Feature Importance Analysis

Random Forest Regressor was used to determine feature importance.

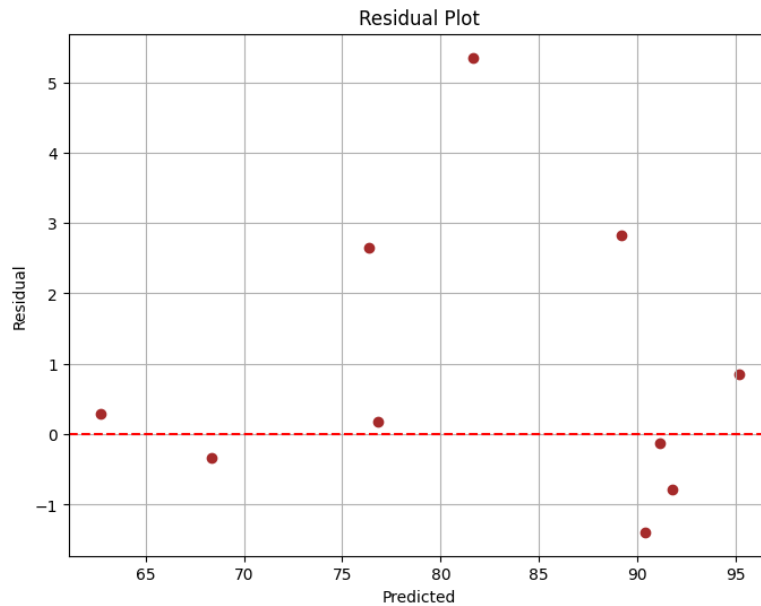
Important factors affecting battery health include:

- Charging Cycles
- Battery Age
- Daily Usage Hours
- Overheating Issues
- Performance Rating



12. Residual Analysis

Residual plots were generated to evaluate prediction errors.



13. Results and Discussion

Among all algorithms tested, Random Forest Regressor produced the highest prediction performance with the best R^2 score and the lowest prediction error. The ensemble nature of Random Forest enabled it to capture complex relationships within the dataset more effectively than other models.

Linear Regression performed adequately but struggled with non-linear relationships. Decision Tree and KNN achieved moderate performance. SVR provided stable predictions but was less accurate than Random Forest.

The visualizations clearly demonstrated the influence of battery age, charging cycles, and usage behavior on battery health.

14. Conclusion

This project successfully implemented and compared multiple machine learning algorithms for laptop battery health prediction. Advanced Matplotlib visualizations were utilized to analyze the dataset and evaluate model performance.

The comparative study revealed that Random Forest Regressor achieved the highest prediction accuracy and demonstrated superior performance compared to Linear Regression, Decision Tree, SVR, and KNN models.

The developed system can assist users in monitoring battery condition and predicting battery degradation, thereby supporting proactive battery maintenance and improved device performance.

15. Future Scope

Future enhancements may include:

- Integration with real-time battery monitoring systems.
- Deep Learning-based battery health prediction.
- Mobile application deployment.
- Cloud-based battery analytics dashboard.
- Predictive maintenance recommendation system.

16. References

1. Scikit-Learn Documentation.
2. Pandas Documentation.
3. Matplotlib Documentation.
4. Kaggle Laptop Battery Health Dataset.
5. Machine Learning for Predictive Analytics Research Papers.